

# Job Seeker Copilot

Engineering a multi-service job-search, evidence and application platform—from initial architecture to realistic failure recovery.

- Bernard McGeever
- Software Developer
- Controlled private beta
- Detailed edition · v2

SearchApplicationsDocuments

REAL PROVIDERSJob search, matching and details

Job MatchesReal providers

SortMost relevantFilterRefresh

Software Developer (188)Software Engineer (Not searched)Full-Stack Developer (Not searched)

Java Developer (Not searched)Backend Developer (Not searched)

SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer

Vermillion Analytics

LondonGBP 35,000 - 45,000 per yearPosted Jun 8, 2026

REED.CO.UKDOCUMENTS PREPARED

Lead Full Stack Software Developer

Spectrum IT Recruitment

GuildfordGBP 80,000 - 100,000 per yearPosted May 8, 2026

REED.CO.UKNOT SAVED

Junior Software Engineer

Harnham - Data & Analytics Recruitment

LondonGBP 30,000 - 40,000 per yearPosted Aug 7, 2026

REED.CO.UKDOCUMENTS PREPARED

Software Engineer (C/C+)

Rise Technical Recruitment Limited

REED.CO.UKDOCUMENTS PREPARED

NOT ENABLED FOR THIS BETA

AI Credit & Payments

Wallet balance and purchasing will be enabled after the secure payment integration is approved.

Open AI Credit area

AVAILABLEReporting & job-search evidence

Your job search progress

29 hours remaining this week.

Refresh

Applications6

Generated documents6100%

Applied00%

Interviews00%

Offers00%

Unsuccessful00%

Recent activity

11/08 20:36Documents

A source- and history-backed case study for recruiters, engineering interviewers and potential partners. Private repositories can be reviewed by arrangement.

# Two reading modes, one evidence base

Pages 1–5 explain the product, significance and professional contribution quickly. The remaining chapters show how services, contracts, persistence, providers, documents, AI controls and test environments actually work.

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**Evidence boundary.** Present-day claims were checked against source, contracts and the running environment. Historical claims use Git/PR/issue evidence. Dated reports prove only their named conditions.

# One workspace for the fragmented work of finding a job

Job Seeker Copilot helps a candidate organise career evidence, discover vacancies, prepare grounded documents, track applications and retain an auditable record of what was used.

It is not “AI writes a CV”. The product is a traceable workflow from verified career evidence and a real vacancy to an application-specific document set.

**Candidate problem**  
Vacancies, versions, portals, evidence and follow-up are scattered across tools.

**Product response**  
Structured profile, multi-provider discovery, document lineage and application tracking.

**Engineering response**  
Owned domain state, provider gateways, versioned contracts and recoverable workflows.

**Current boundary**  
Controlled private beta; local persistent operation, not public production.

|                                                  |                                                    |                                     |                      |
|--------------------------------------------------|----------------------------------------------------|-------------------------------------|----------------------|
| 31                                               | 28                                                 | 7                                   | 5                    |
| repositories audited—not used as a vanity metric | healthy application services in the manual profile | independent PostgreSQL state owners | job-provider domains |



# Full-stack delivery, integration and product ownership

## Product and frontend

- ✓ Progressive account, profile and evidence UX
- ✓ Search, provider, failure and empty states
- ✓ Applications, documents, history and reporting
- ✓ Accessible browser journeys

## Java/Spring platform

- ✓ Domain services and gateways
- ✓ REST/OpenAPI contracts
- ✓ PostgreSQL/Flyway ownership
- ✓ Idempotent multi-service workflows

## AI/document engineering

- ✓ Evidence snapshots and prompts
- ✓ Parsing, repair and grounding
- ✓ Recovery and deterministic fallback
- ✓ DOCX/PDF export and lineage

## External integration

- ✓ Five job sources
- ✓ Google/Postcodes location
- ✓ OpenAI provider boundary
- ✓ Stripe architecture, disabled browser path

## Confidence engineering

- ✓ System Data and canonical personas
- ✓ Playwright/Cucumber E2E
- ✓ Live/manual validation
- ✓ Capacity workloads and cost evidence

## Delivery system

- ✓ Compose profiles and persistent volumes
- ✓ Multi-repository PR/CI discipline
- ✓ Contract generation and revision locks
- ✓ Central documentation and audits

**Accurate professional claim.** Bernard conceived the product, set requirements, directed AI-assisted implementation, made and reviewed architecture decisions, integrated the services, evaluated real behaviour and repeatedly drove defects to resolution. The evidence is delivery judgement—not a claim that every line was typed without modern tooling.

## Not a prototype claim

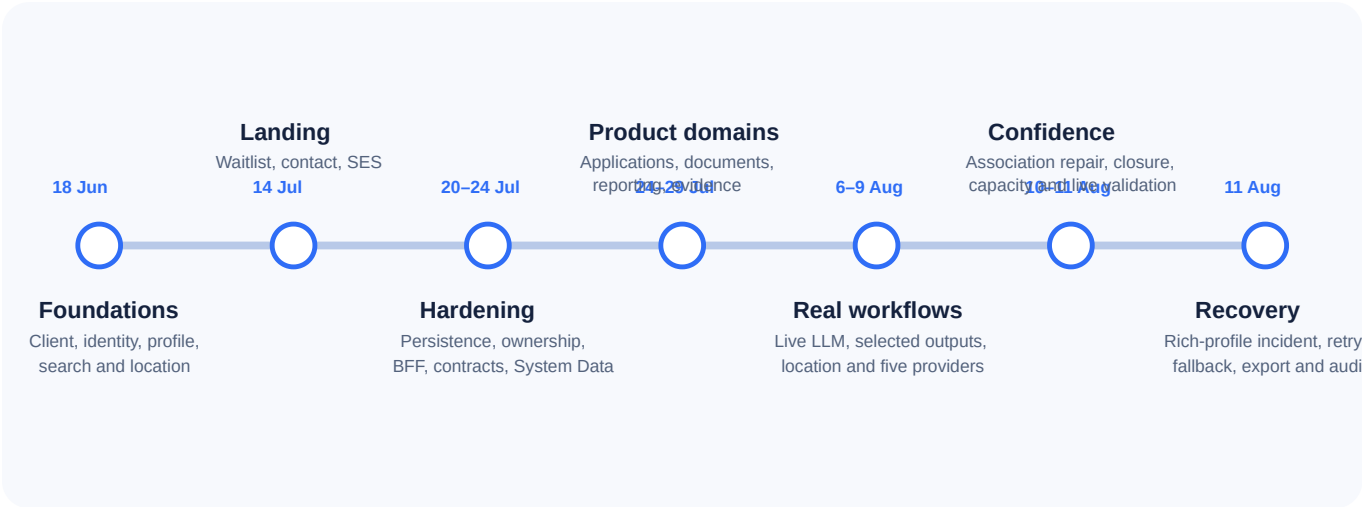
Persistent state, identity boundaries, real contracts, provider adapters, lifecycle rules and failure recovery exist behind the UI.



## What this demonstrates

Problem decomposition, full-stack engineering, debugging, integration, testing, documentation and product ownership.

# A concentrated sequence of architectural maturation



|              |                                                                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18 June      | Initial client, authentication, profile, location, job search and provider repositories establish the application foundation.                                                                      |
| 14–22 July   | Public landing/waitlist work is followed by identity hardening, durable migrations, HttpOnly session architecture, vulnerability gates and accessible flows.                                       |
| 22–24 July   | System Data and Playwright/Cucumber are imported; generated client binaries are removed; producer-owned contracts and clean builds become policy.                                                  |
| 24–29 July   | Application tracking, document storage/export, reporting, payments, evidence revisions, snapshots and claim provenance become separate owned domains.                                              |
| 6–9 August   | Real OpenAI paths, retained generation, exact document selection, uploads, Location Service/Google isolation and five-source discovery are integrated.                                             |
| 10–11 August | Realistic E2E exposes application/document association gaps; private-beta closure, live providers and capacity validation follow.                                                                  |
| 11 August    | A rich real profile exposes duplicate structure and output/deadline weaknesses. Repair, bounded retry, unknown-outcome reconciliation, deterministic fallback and export quality are strengthened. |

**History boundary.** The repositories visible here were migrated or created in June–August 2026; this timeline describes evidenced repository history, not undocumented work before the recorded baselines.

# Every screen is backed by an ownership decision

Jobseeker Copilot

Create your account, then shape a job search that works for you.

Create account

Sign in

Your account details

No CV, work history, skills or postcode required.

Full name

Use between 1 and 100 characters.

Email address

Password

Use 8 to 100 characters. Spaces are not allowed.

Create account

Registration. Minimal first entry; no CV or postcode required. Browser tokens remain in HttpOnly cookies.

Software Developer (188)

Software Engineer (Not searched)

Full-Stack Developer (Not searched)

Java Developer (Not searched)

Backend Developer (Not searched)

SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER - MORE AVAILABLE

Software Engineer

REED.CO.UK

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Software Engineer (C/C+)

REED.CO.UK

DOCUMENTS PREPARED

Rise Technical Recruitment Limited

📍 Stevenage 💰 Salary not specified 📅 Posted date unavailable

Software Developer Trainee

REED.CO.UK

NOT SAVED

ITOL Recruit

📍 E62RP 💰 GBP 28,000 - 35,000 per year 📅 Posted date unavailable

Lead AI Software Engineer

REED.CO.UK

NOT SAVED

Avanti

📍 W60QU 💰 GBP 80,000 - 120,000 per year 📅 Posted date unavailable

Discovery. Provider status, source attribution and application state are visible rather than hidden.

Account

→

Profile/evidence

→

Search/detail

→

Save

→

Select evidence

→

Generate/upload

→

Track/report

| User step        | Backend engineering                                                | Durable truth             |
|------------------|--------------------------------------------------------------------|---------------------------|
| Register/sign in | BFF cookie and CSRF boundary; UMG; RS256/JWKS                      | Authentication DB         |
| Build profile    | Progressive sections, revisions, confirmation and provenance       | Profile DB                |
| Discover/save    | Five-provider fan-out, mapping, dedupe and snapshot                | Job DB                    |
| Generate         | Evidence snapshot, wallet reservation, LLM validation and fallback | Generation ledger + Store |
| Apply/track      | Exact document selections, immutable events and transitions        | Application DB            |
| Report           | Read-only projections from owner-scoped services                   | Upstream owners           |

Product journey

06

# Decomposed by ownership, corrected by failure

## One authoritative owner

Accounts, profile evidence, saved vacancies, applications, documents, wallet entries and operation checkpoints have explicit systems of record.

## Gateways isolate trust

Browser, provider and model DTOs/credentials stop at controlled boundaries. Domain services consume canonical contracts.

## Immutable references

Evidence snapshots, saved jobs, document versions and applied selections resist later mutation.

## Failure is state

Retries, ambiguous outcomes and partial side effects are persisted and reconciled rather than hidden behind a generic error.

## Evolution pattern

### Original simplification

Generate content, save output and continue synchronously.



### Current operation

Persist intent, snapshot inputs, reserve credit, checkpoint side effects, validate output, approve exact versions and reconcile application references.

### Why microservices?

Different trust, state and failure boundaries—not repository count for its own sake.

### Why contracts?

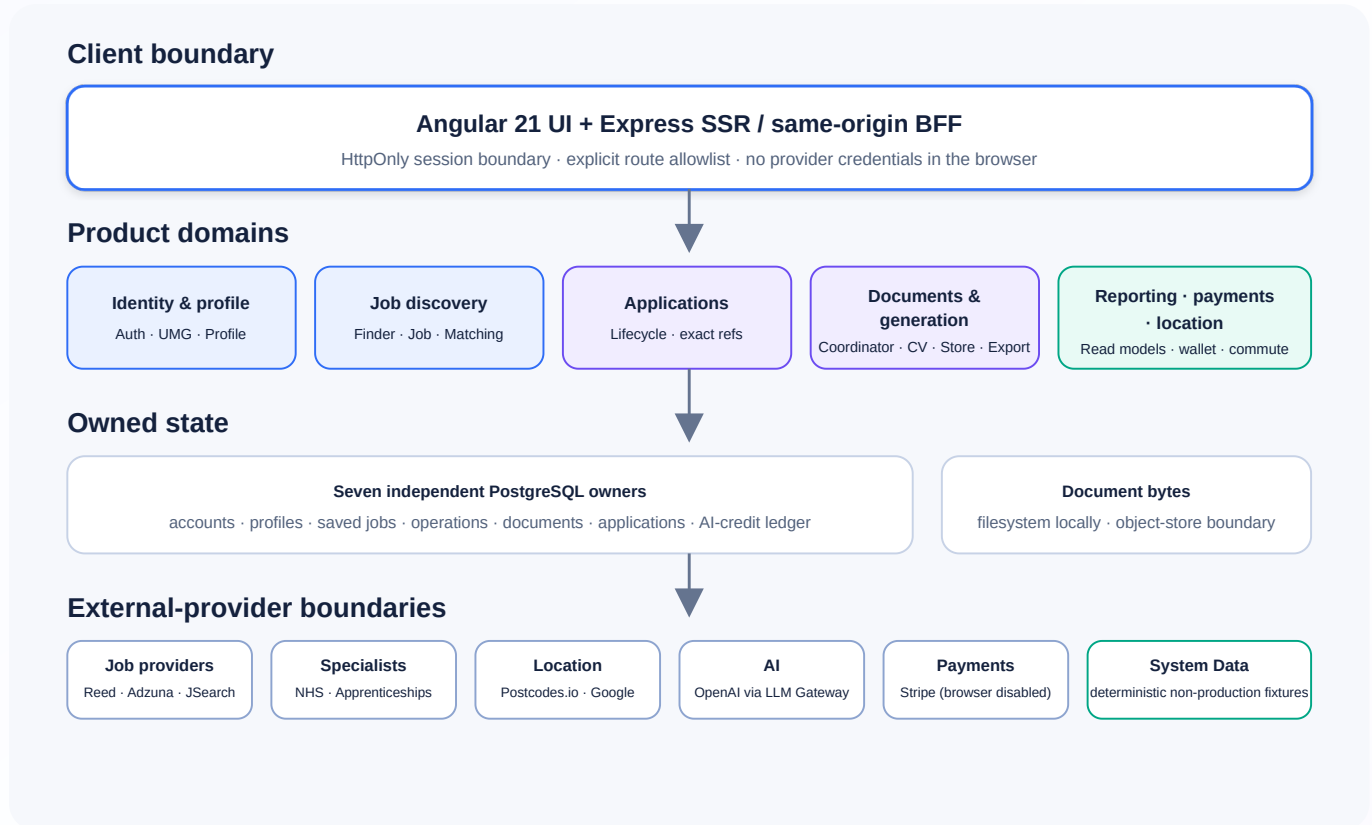
Independent repositories otherwise compile against yesterday's assumptions.

### Why fixtures?

Repeatable failure and zero-spend regression, kept distinct from live confidence.

Architecture was not fixed in advance. Document association, rich evidence, provider inconsistency and realistic concurrency each forced clearer ownership or recovery rules.

# Major boundaries without a 30-service spaghetti diagram



## Browser boundary

Angular + Express BFF, allowlisted routes, HttpOnly cookies and CSRF.

## Owned state

Seven PostgreSQL owners plus document bytes in an object-provider boundary.

## External boundaries

Five job sources, OpenAI, Google, Postcodes.io and disabled-by-default Stripe.

Arrows show material runtime calls, not every health, lifecycle or fixture-control edge. The full catalogue appears on pages 40–42.

# Complexity is meaningful only when it explains the engineering

|                                                       |                                                               |                                                                        |                                                                                      |
|-------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <div>31</div> <div>workspace repositories</div>       | <div>26</div> <div>Java/Spring application repositories</div> | <div>1+1</div> <div>authenticated Angular app + separate landing</div> | <div>7</div> <div>PostgreSQL state owners</div>                                      |
| <div>9</div> <div>external provider/API domains</div> | <div>6</div> <div>supported Compose profiles</div>            | <div>32/244</div> <div>retained regression scenarios/steps</div>       | <div>4</div> <div>document artifacts per successful pair: CV/letter × DOCX/PDF</div> |

| Domain                 | Repositories/components               | Why separation matters                                                                   |
|------------------------|---------------------------------------|------------------------------------------------------------------------------------------|
| Identity/profile       | Authentication, UMG, User Profile     | Credentials, browser sessions and career evidence have different access/lifecycle rules. |
| Discovery              | Finder, Job, Matching, five providers | Provider failures and DTOs remain outside the canonical product model.                   |
| Documents/AI           | Generation, CV, LLM, Store, Export    | Orchestration, untrusted model output, immutable truth and rendering are independent.    |
| Applications/reporting | Tracker, Reporting/Gateway            | Read models cannot silently mutate lifecycle truth.                                      |
| Payments               | Payment, Gateway, Stripe              | Wallet accounting is not delegated to the model or checkout provider.                    |
| Location               | Gateway, Service, Google, Postcodes   | Persistable location truth differs from transient provider/session data.                 |

**Not a scale claim.** Repository/service counts show the contract and integration problem Bernard solved. They do not prove user adoption, high availability or production throughput.

# The Angular application is a product boundary, not a thin skin

## Major user areas

- ✓ Registration, sign-in, refresh and return
- ✓ Dashboard, profile, preferences and evidence manager
- ✓ Search, provider status, job details and commute
- ✓ Evidence selection and asynchronous generation
- ✓ Documents, versions, archive/recovery and downloads
- ✓ Applications, exact selections, status and reporting

## Engineering behaviour

- ✓ Generated/typed API models where contracts exist
- ✓ Same-origin Express route allowlist
- ✓ HttpOnly session cookies; no bearer token in localStorage
- ✓ CSRF on state-changing proxy routes
- ✓ Provider URLs/content treated as untrusted
- ✓ Loading, partial, empty, expired and recoverable states

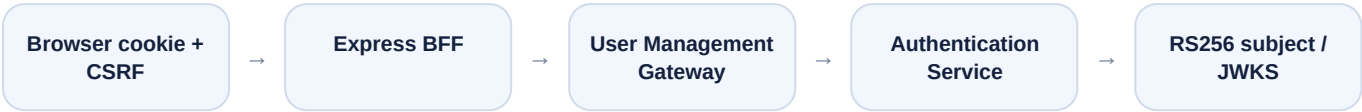
```
app.post('/api/auth/refresh', async (req, res) => {
  await proxyUserManagementRequest('/api/auth/refresh', 'POST', req, res, true);
});
if (forwardSessionCookies && result.setCookies.length) {
  res.setHeader('Set-Cookie', result.setCookies);
}
registerDocumentGenerationRoutes(app, {
  accessCookieName, csrfCookieName, origin, timeoutMs
});
```

**Why the BFF exists.** Browser code never chooses arbitrary service destinations or stores access tokens. The server boundary converts cookies into controlled downstream requests and stable, redacted failures.

The dashboard is divided into three main sections: Search, Applications, and Documents. The Search section displays job matches for 'Software Developer' with filters for 'Real providers' and 'Sort: Most relevant'. It lists several job openings, including 'Software Engineer' at Vermillion Analytics, 'Lead Full Stack Software Developer' at Spectrum IT Recruitment, 'Junior Software Engineer' at Harnham - Data & Analytics Recruitment, 'Software Engineer (C/C++)' at Rise Technical Recruitment Limited, and 'Software Developer Trainee' at ITOL Recruit. The Applications section shows a progress bar for 'Your job search progress' with 29 hours remaining this week. The Documents section shows a list of recent activity, including 'Downloaded previous cover letter version 1' and 'Downloaded previous CV version 1'.

The dashboard composes application, document and reporting state from separate owners without becoming their database.

# Identity is derived, forwarded and independently enforced



**Authentication owns**

- canonical email identity and password state;
- registration, login, refresh, logout and reset;
- refresh/session lifecycle in PostgreSQL;
- asymmetric token signing and public JWKS.

**Consumers enforce**

- validated token subject, never caller-supplied user ID;
- route-scoped service identities;
- owner-scoped reads and writes;
- redacted browser-safe error contracts.

## Architecture change

**Earlier risk**

Bearer/session material in browser storage and caller-controlled downstream identity widened the impersonation boundary.

→

**Current boundary**

HttpOnly cookies terminate at the same-origin BFF; gateways/services derive and verify the subject independently.

| Control                | Evidence                                                   | Limit                                                            |
|------------------------|------------------------------------------------------------|------------------------------------------------------------------|
| Credential persistence | Migration-backed PostgreSQL                                | Local/private-beta operation is not a security certification.    |
| Abuse resistance       | Bounded login attempts, input normalisation, stable errors | Production threat monitoring still requires deployed operations. |
| Lifecycle              | Refresh/logout/reset and account cleanup calls             | Cross-service retention policy remains ongoing product work.     |

Evidence: Authentication PRs #16/#18/#24; Client PRs #15–#18; User Management contract/security PRs #18–#24.

# Career history is structured product data

Synthetic fixture: progressive employment, projects, qualifications, skills and preferences.

Synthetic fixture: evidence entries expose category, confirmation, provenance and revision state.

**Entry**

Stable category and identity; visibility, lifecycle and review state.

**Revision**

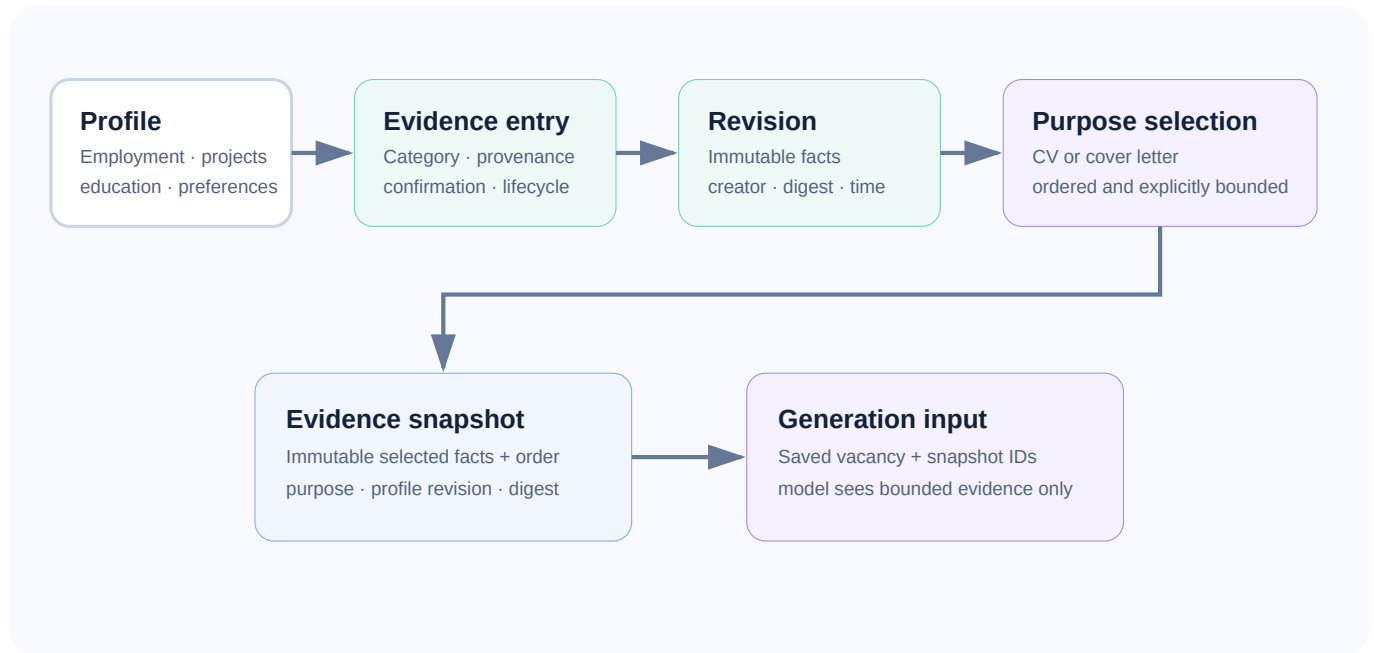
Immutable facts, creator, confirmation and content digest.

**Snapshot**

Purpose, selected revisions, section order, profile revision and digest.

**Why this exists.** Repeatedly pasting a free-text biography into a model makes provenance and later correction impossible. Structured evidence allows a user to confirm facts once, select them for a purpose and retain exactly what supported a generated document.

# Selection becomes an immutable generation input



```

EvidenceSnapshot create(userId, request) {
  require(profile.revisionId && profile.contentDigest);
  for (entryId : request.entryIds) {
    revision = eligibleUserConfirmedRevision(entryId, userId);
    enforceSectionOrderAndFactLimit(revision);
    snapshot.addSelection(copyFactsAndDigests(revision));
  }
  snapshot.snapshotDigest = digest(snapshot);
  return repository.saveAndFlush(snapshot);
}

```

## CV purpose

Allows broad but bounded career evidence and selected section order.

## Cover-letter purpose

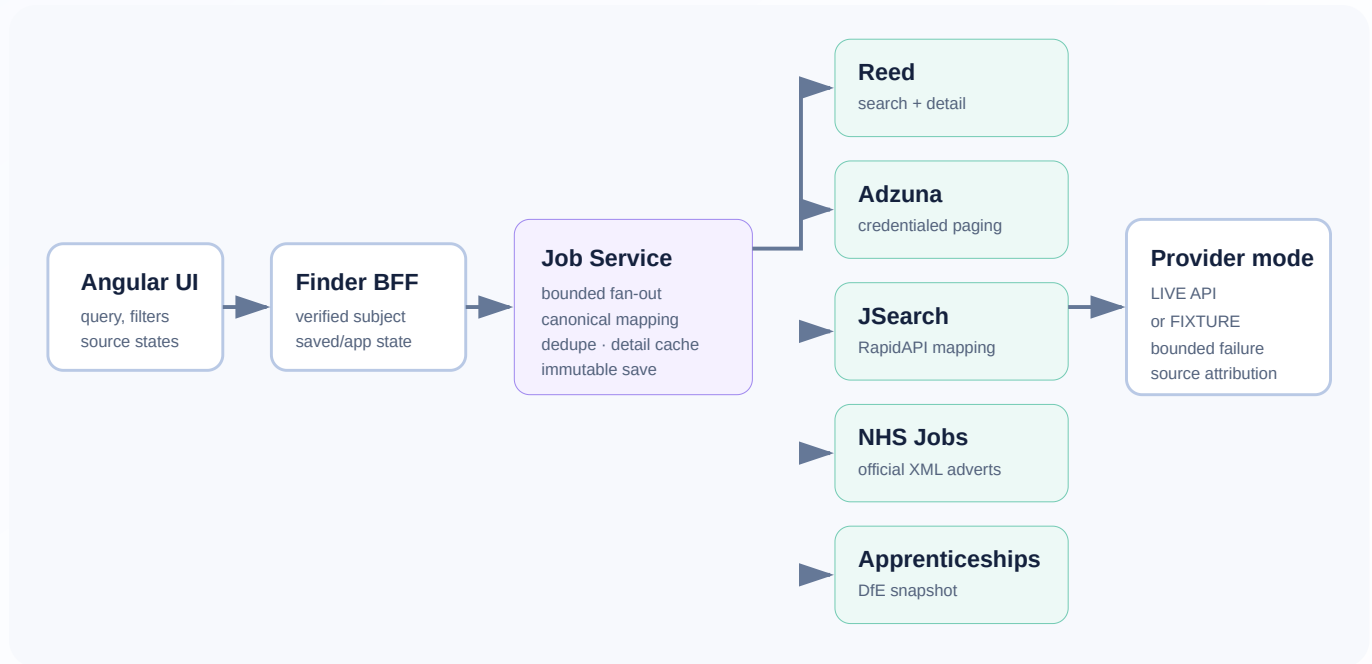
Can use a narrower, vacancy-specific evidence subset.

## Application provenance

Snapshot and claim-ledger IDs/digests travel with document versions.

**Boundaries.** Duplicate selections, unconfirmed/latest-invalid revisions, hidden/review-required evidence and oversized snapshots fail before the model call.

# Five providers become one truthful product response



## Fan-out

Provider calls are bounded independently. One failure can produce a partial response rather than erase healthy sources.

## Mapping

Adapters convert provider IDs, dates, salary, workplace type, descriptions and source URLs.

## Save

The selected result is hydrated where supported and persisted as an immutable vacancy snapshot.

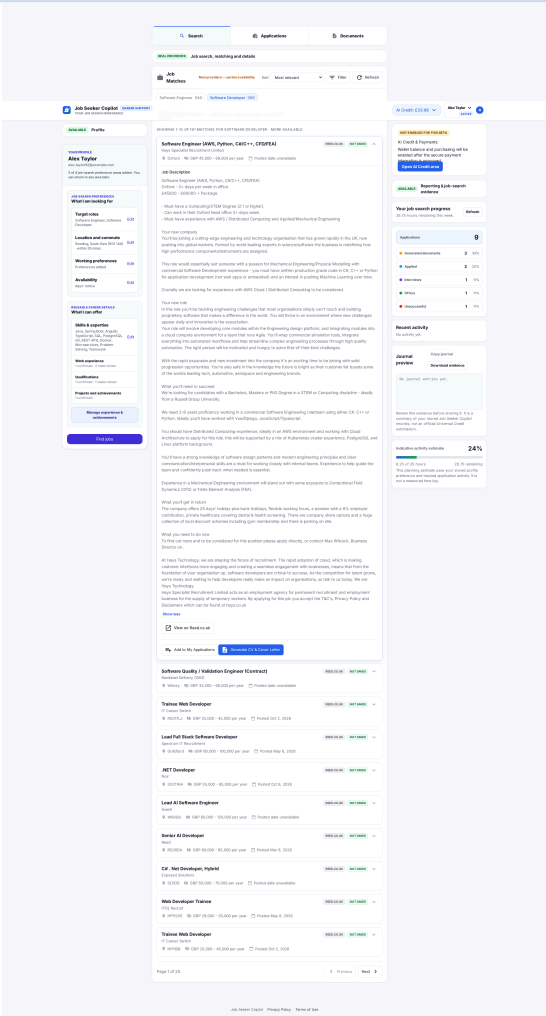
The screenshot shows a web interface for job discovery. At the top, there are filters for 'Software Developer' (188), 'Software Engineer' (Not searched), and 'Full-Stack Developer' (Not searched). Below the filters, it says 'SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER - MORE AVAILABLE'. The main content area displays a list of job listings, each with a title, company, location, salary range, and posted date. Each listing has a 'REED.CO.UK' button and a 'DOCUMENTS PREPARED' or 'NOT SAVED' status. The listings include:

- Software Engineer** at Vermillion Analytics, London, GBP 35,000 - 45,000 per year, Posted Jun 8, 2026.
- Lead Full Stack Software Developer** at Spectrum IT Recruitment, Guildford, GBP 80,000 - 100,000 per year, Posted May 8, 2026.
- Junior Software Engineer** at Harnham - Data & Analytics Recruitment, London, GBP 30,000 - 40,000 per year, Posted Aug 7, 2026.
- Software Engineer (C/C++)** at Rise Technical Recruitment Limited, Stevenage, Salary not specified, Posted date unavailable.
- Software Developer Trainee** at ITOL Recruit, E62RP, GBP 28,000 - 35,000 per year, Posted date unavailable.
- Lead AI Software Engineer** at Avant, W60QU, GBP 80,000 - 120,000 per year, Posted date unavailable.

Point-in-time live-provider evidence. Changing external adverts and provider uptime are not regression assertions.

# Each source has different contracts and failure shapes

| Provider        | Authentication/acquisition            | Mapping/detail behaviour                                             | Modes and known limits                                              |
|-----------------|---------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|
| Reed            | API key; bounded search/detail calls  | Dedicated detail endpoint, source/apply URLs, salary and posted data | Fixture default; bounded live sample; advert text can be incomplete |
| Adzuna          | App ID/key; paged search              | Credentialed pagination, workplace type and canonical mapping        | Fixture default; paging/quota and provider date differences         |
| JSearch         | RapidAPI key/host                     | Provider records mapped from aggregated upstream sources             | Fixture default; IDs, descriptions and dates vary                   |
| NHS Jobs        | Official self-serve XML path          | Authoritative NHS identity and advert/source provenance              | Live path plus deterministic in-service fixture                     |
| Apprenticeships | DfE Display Advert API v2 credentials | Authenticated acquisition to atomic snapshot; training details       | Readers see last complete snapshot, never half-refresh state        |



Synthetic fixture: expanded advert, source link and generation controls. Provider text is displayed as untrusted content.

Engineering work beyond “calling an API”. Authentication, paging, XML/JSON shape, missing fields, status/date semantics, source identity, provider errors, attribution, terms-safe caching and deterministic equivalents all differ.

# Heterogeneous adverts need explainable identity

## Canonical fields retained

- ✓ canonical job ID and provider/external IDs;
- ✓ all source references and canonicalised URLs;
- ✓ title, company, location and workplace type;
- ✓ salary, posted/expiry dates and description;
- ✓ specialist/apprenticeship details and provenance.

## Duplicate signals

- ✓ same canonical apply URL;
- ✓ overlapping source/apply options;
- ✓ normalised company + high title similarity;
- ✓ normalised location + compatible dates;
- ✓ compatible descriptions within configured thresholds.

```
if (sameCanonicalApplyUrl(left, right) || overlappingApplyOptions(left, right)) return 1.0;
if (sameCompany(left, right)
    && titleSimilarity(left, right) >= 0.90
    && sameLocation(left, right)
    && postedDatesCompatible(left, right)
    && descriptionsCompatible(left, right)) return 1.0;

mergeDuplicateJobs(primary, duplicate); // also merges every source reference
```

### Naïve result

Two apparently identical cards or, worse, one merged record that loses the source needed to apply.



### Current result

One canonical display record can retain multiple provider references, a preferred apply URL and field provenance.

**Known boundary.** Deduplication is deliberately rule-based and explainable. It can still miss semantically identical adverts or avoid merging uncertain ones; it is not an opaque similarity model.

# A saved vacancy becomes owned lifecycle state

Application state, milestones and exact document selection remain visible after generation.

## Identity

Owner + canonical/provider/external job identity; idempotent creation independent of generation.

## Lifecycle

SAVED → DOCUMENTS\_GENERATED → APPLIED → INTERVIEW → OFFER, with guarded transitions.

## History

Immutable events and milestone dates survive refresh and reporting projection.

```
PUT /applications/{id}/document-selections
// CV and cover-letter slots are both mandatory:
// SELECTED(exact approved version) or OMITTED.
// Store verifies owner, type, availability and provenance.
// APPLIED freezes both choices atomically.
```

**Separation of truth.** Document Store proves what a version is and whether it is available. Application Tracker decides which exact versions the candidate selected and whether that choice is now frozen.

Evolution: Tracker PRs #26/#29/#30/#42; Generation PRs #90-#95/#105/#120; E2E application-document selection.

# A document is a family, immutable versions and files —not one mutable blob

Document families  
23  
Each family is counted once

CV families  
17  
All retained versions

Cover-letter families  
6  
All retained versions

All document families 23

CVs 17

Cover letters 6

Search document families

Sort by Recently updated

Document families

Refresh

Cover letter — Software Engineer

Cover letter

1 retained version Latest v1

AI-tailored

Software Engineer — Harnham - Data & Analytics Recruitment

Open for associations

No current version

CV — Software Engineer

CV

1 retained version Latest v1

AI-tailored

Software Engineer — Harnham - Data & Analytics Recruitment

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Software Developer — Reed

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Software Developer — Reed

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Software Developer — Reed

Open for associations

No current version

Families expose type, job scope, source, retention, current state and application associations.

## Document Store owns

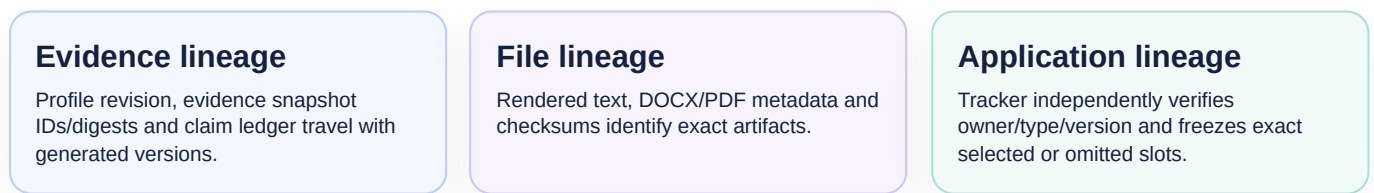
- ✓ family and immutable version records;
- ✓ generated/uploaded source and lineage;
- ✓ current/approved pointers;
- ✓ DOCX/PDF metadata, checksum and MIME;
- ✓ filesystem bytes locally or S3-compatible provider;
- ✓ archive, restore, recoverable delete and tombstones.

## Upload boundary

PDF/DOCX ≤10 MiB; bounded extraction; MIME/signature checks; isolated ClamAV scanning; no model or AI-credit call.

| Lifecycle action | Meaning                       | What it does not do                 |
|------------------|-------------------------------|-------------------------------------|
| Set current      | Select family recommendation  | Does not rewrite an application     |
| Approve          | Make exact version eligible   | Does not auto-select it for a job   |
| Archive/restore  | Reversible organisation state | Does not mutate bytes               |
| Delete/recover   | Recovery-window lifecycle     | Purge remains separately controlled |

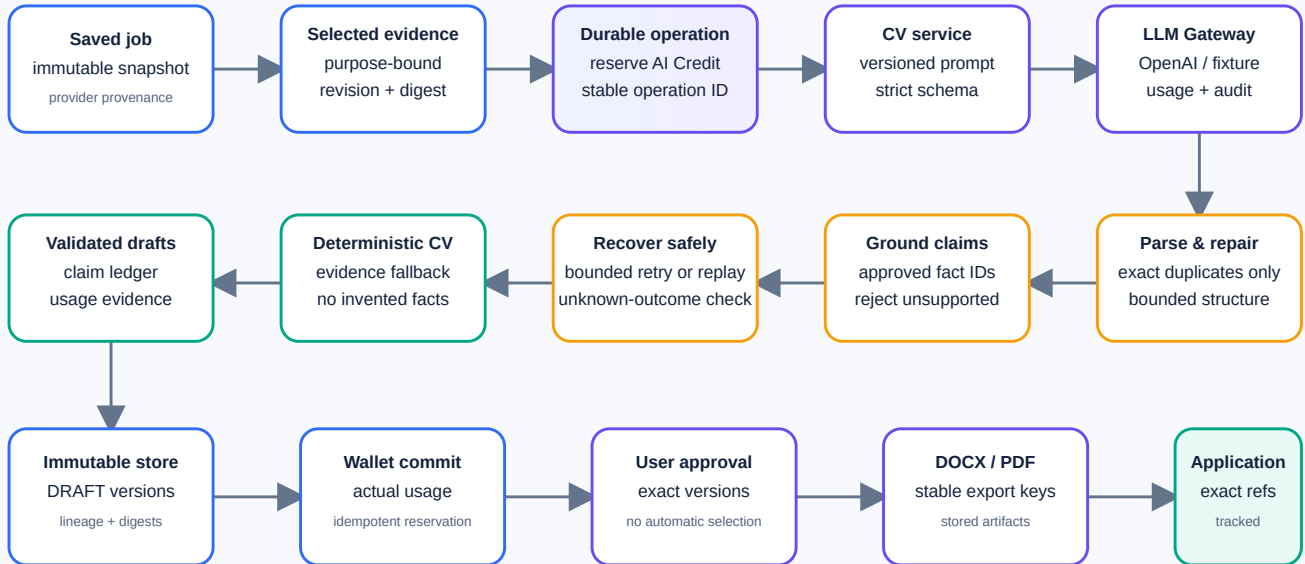
**Historical problem solved.** Regenerating a CV tomorrow must not silently change the version associated with yesterday's application.



## Version and application truth

# Generation is a durable product operation

The LLM is a bounded collaborator, not a source of truth



## Before the model

Persist operation/deadline; resolve saved vacancy; create purpose snapshots; estimate and reserve credit.

## After the model

Validate claims; store draft versions; commit actual usage; wait for explicit approval.

## After approval

Set pointers; export exact versions; create/reconcile application and exact document selections.

### Software Engineer

Vermillion Analytics

London GBP 35,000 - 45,000 per year Posted Jun 8, 2026

#### Job Description

Software Engineer Location: Hybrid / Remote (UK-based) Team: Engineering Salary: Up to 45000 Reports to: VP of Engineering You must be UK based and eligible to work. About Our Client Our client is one of Europe's fastest-growing software companies, on a mission to solve productivity problems for people-intensive industries using AI and automation. Since being founded, they've focused on automating the manual, tedious parts of complex workflows...

[Read full advert](#)

[View on Reed.co.uk](#)

✓ CV and Cover Letter generated successfully

#### CV

[Download DOCX](#)

[Download PDF](#)

[Upload CV](#)

#### COVER LETTER

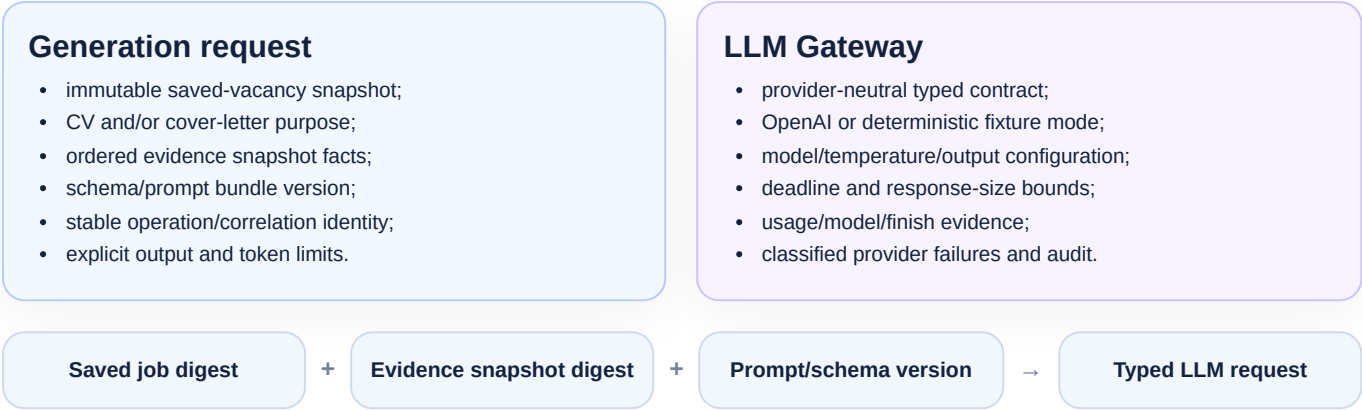
[Download DOCX](#)

[Download PDF](#)

[Upload Letter](#)

Real vacancy and provider path; exact generated drafts expose DOCX/PDF only after the operation completes.

# The model receives a bounded task, not the whole application



## Trusted identity stays outside the prompt

Contact details are rendered from trusted profile fields. Provider advert text and prior documents remain untrusted content. The model cannot grant itself new evidence identifiers or change the user identity.

| Boundary      | Failure handled                         | Evidence retained                         |
|---------------|-----------------------------------------|-------------------------------------------|
| Admission     | Oversized/unsupported request           | No paid call                              |
| Provider      | Timeout, rejection, limit, network      | Model, usage, finish/error classification |
| Domain parser | Malformed/overlong structure            | Bounded diagnostic/replay metadata        |
| Grounding     | Unsupported or incorrectly mapped claim | Claim/evidence path failure               |

A successful HTTP response is not a successful CV. The domain accepts content only after parsing, structural policy and evidence validation.

# Fluent output is treated as untrusted data



```
for (claim : generatedClaims) {
  requireKnownContentPath(claim.contentPath());
  requireApprovedEvidenceIds(claim.evidenceIds());
  requireClaimSupportedBySnapshot(claim, evidenceCatalog);
  rejectSpecificOrSensitiveClaimWithoutExactEvidence(claim);
}
validatedLedger = digest(claims, snapshot, promptVersion);
```

**Repair boundary.** Removing an exact duplicate bullet cannot create a new fact. Rewriting an unsupported achievement could. The first may be repaired; the second must be rejected or replaced by grounded fallback.

|                                                          |                       |                        |                                    |
|----------------------------------------------------------|-----------------------|------------------------|------------------------------------|
| 17                                                       | 80,012                | 23,419                 | \$0.069487                         |
| bounded live calls in the closure<br>quality/cost sample | input tokens observed | output tokens observed | observed sample cost, not forecast |

# “Try again” is unsafe when money or side effects may already exist

| Failure/state                   | Meaning                                                          | Safe behaviour                                                  |
|---------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| DOWNSTREAM_RETRYABLE            | Replay-safe step failed before/after a known idempotent mutation | Resume bounded step with same key                               |
| MODEL_REJECTED / invalid output | Provider responded but content failed policy                     | Narrow repair, retained replay or fallback                      |
| GENERATION_OUTCOME_UNKNOWN      | Provider invocation may have completed but response is uncertain | Reconcile retained outcome; do not immediately call model again |
| DEADLINE_EXCEEDED               | Absolute operation deadline elapsed                              | Checkpoint and resume only replay-safe phases                   |
| LIMIT_REACHED                   | Model/output/admission bound reached                             | Fail with actionable classification or fallback where eligible  |

```
generationOutcomeUnknown(operation, lease, exception) {
  return checkpoint(operation, lease,
    GENERATION_OUTCOME_UNKNOWN,
    "GENERATION_OUTCOME_UNKNOWN",
    "provider outcome is ambiguous; automatic retry is disabled " +
    "to prevent duplicate cost");
}
```

## Stable keys

Reservation, generation, storage, export and application side effects derive from one operation ID.

## Retained response

A captured provider response can be re-parsed/revalidated without another paid call.

## Leases/deadlines

Concurrent workers coalesce; persisted absolute deadlines prevent infinite recovery.

**Wallet rule.** Release only when model use is known not to have completed. Commit actual usage idempotently. An ordinary browser retry must never become a second user charge.

# Realistic data broke assumptions that compact fixtures did not

- 1 **Failure:** a real rich profile produced exact duplicate structured CV content.  
The model response was fluent, but strict validation correctly rejected the repeated structure.
- 2 **Failure:** a subsequent generation reached the previous output/deadline boundary.  
The operation could not safely assume whether another paid request was appropriate.
- 3 **Why confidence was overstated:** compact deterministic personas exercised contracts but not realistic evidence volume, role ordering or output length.

## Architecture response

- ✓ purpose-appropriate CV output budget;
- ✓ exact-duplicate structural repair;
- ✓ re-run full claim validation after repair;
- ✓ stronger vacancy/evidence selection and role prominence.

- ✓ bounded safe LLM retry classification;
- ✓ unknown-outcome reconciliation;
- ✓ selected-output-aware recovery;
- ✓ deterministic evidence-based CV fallback.

### Old product outcome

Model rejection surfaced as a failed generation operation even when safe evidence existed.



### Current product outcome

The model result may still be rejected, while the operation can complete with a grounded deterministic CV and the cover letter can remain the accepted model output.

**Engineering lesson.** Deterministic fixtures remain essential for regression, but representative rich profiles and paid/live boundaries are separate confidence layers.

Evidence: CV PRs #69/#71; LLM #27; Generation #122; Export #21; client partial-result PR #103.

# Fail useful without pretending the model succeeded

```
/** Conservative CV from approved evidence only.
 * No semantic embellishment: canonical fields and narrative bullets
 * are copied from evidence; ordering uses lexical vacancy overlap. */
cv.targetRole = input.job.title;
addSkills(cv, rankedApprovedSkills.limit(12));
addEmployment(cv, approvedEmployment.limit(12));
addQualifications(cv, approvedQualifications.limit(8));
addProjects(cv, approvedProjects.limit(4));
notes.tailoringSummary = "evidence-based CV produced because model generation
did not yield an acceptable draft";
```

## What fallback can do

- ✓ copy canonical job/profile/evidence fields;
- ✓ rank approved skills and evidence by lexical vacancy overlap;
- ✓ preserve dates, organisations and fact provenance;
- ✓ produce the same typed CV schema and claim ledger;
- ✓ continue through Store, export and application workflows.

## What fallback must not do

- invent seniority, metrics or qualifications;
- infer unstated technologies;
- rewrite uncertain dates;
- claim semantic understanding or model quality;
- silently masquerade as AI-tailored output.

**Product value.** The operation remains useful even when the probabilistic collaborator is unavailable or unacceptable. Safety and completion are no longer identical to “the model returned JSON”.

---

# 12

maximum fallback skills

---

# 12

maximum employment groups

---

# 8

maximum qualifications

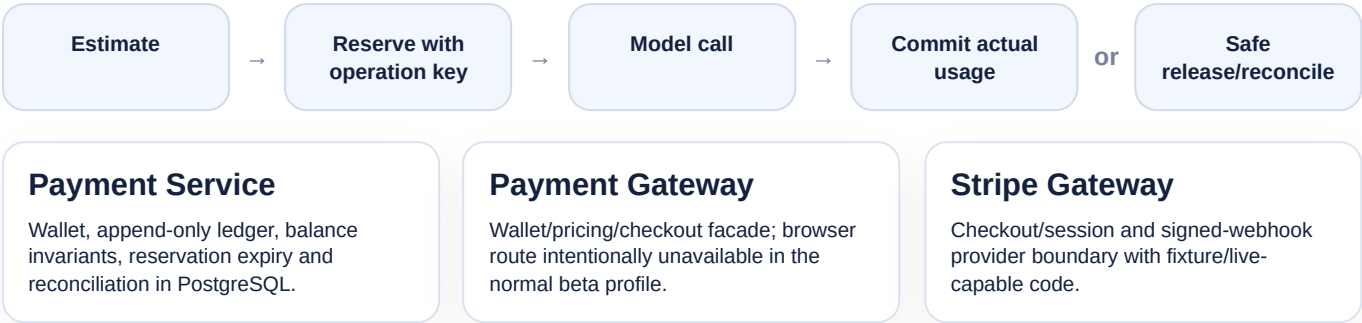
---

# 4

maximum projects



# Substantial accounting architecture, even with live checkout disabled



| Reservation state | Allowed meaning                                          | Retry property                                         |
|-------------------|----------------------------------------------------------|--------------------------------------------------------|
| RESERVED          | Credit temporarily unavailable while operation is active | Same operation key returns/reconciles same reservation |
| COMMITTED         | Actual model usage charged                               | Repeated commit is idempotent                          |
| RELEASED          | Provider use known not to require charge                 | Stable key cannot create a second reservation          |
| Unknown/recovery  | Outcome needs operator/automatic reconciliation          | No speculative release and re-charge                   |

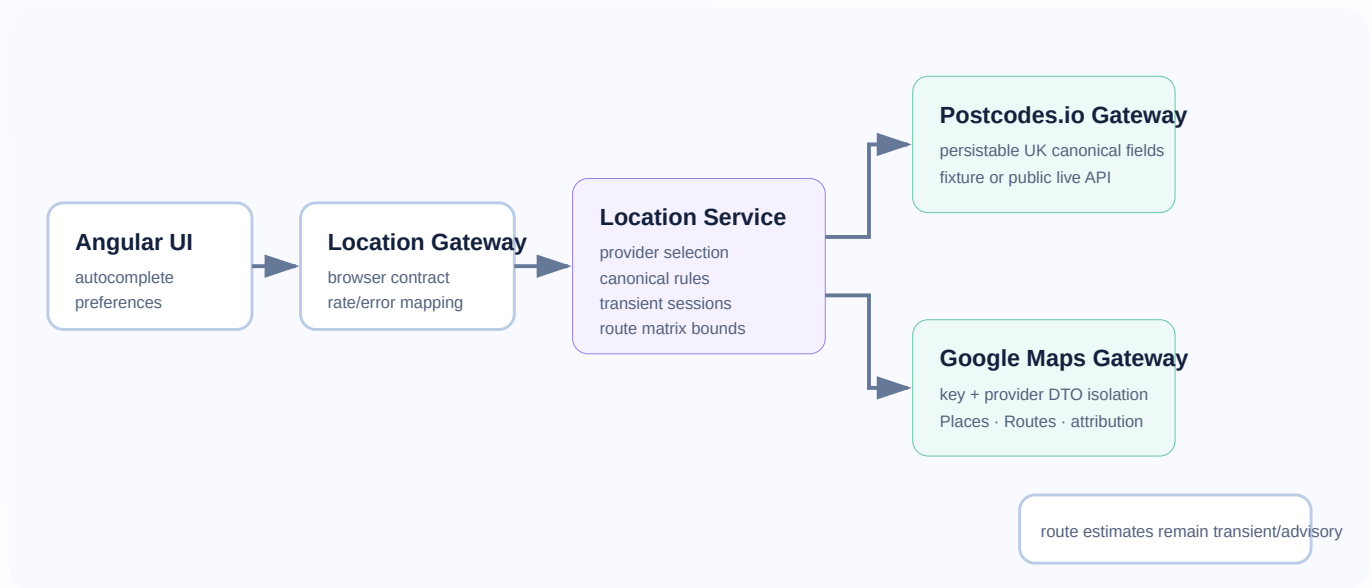
**Implemented**

Wallet, pricing, fixture purchase, reservations, commit/release, operation keys, expiry, recovery and Stripe backend boundary.

**Not exposed**

Ordinary browser live Stripe checkout. Zero live Stripe charges were made in the validated environment.

# Persistable place truth is separated from transient provider data



## Canonical location

Stable label, country, coordinates/postcode authority and provider provenance suitable for profile persistence.

## Autocomplete session

Bounded provider token/session and suggestions remain transient; attribution is required.

## Commute matrix

At most five eligible destinations, selected travel modes and advisory route estimates—never suitability scores.

Location and commute

Reading · within miles

Town or postcode

Reading

5 matching locations found.

Suggestions provided by Google Maps

Reading

Reading Street, Tenterden

Whitley, Reading

Reading Street, Broadstairs

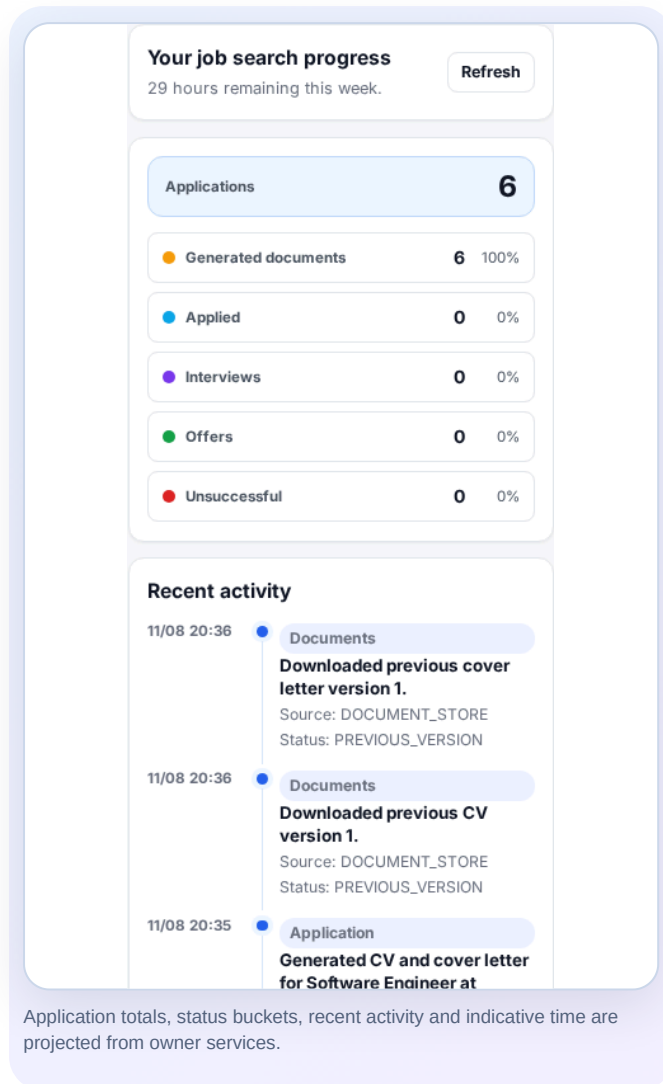
Newtown, Reading

Edit

A real browser request returned five suggestions with required Google Maps attribution. The generic Reading query was not saved.

**Why another service?** The original Location Gateway directly represented Postcodes.io-style operations. Adding Google Places/Routes required a provider-neutral orchestration layer so the browser contract, canonical persistence and provider-specific credentials/DTOs stayed separate.

# Reporting observes domain truth; it does not own or mutate it



## Sources

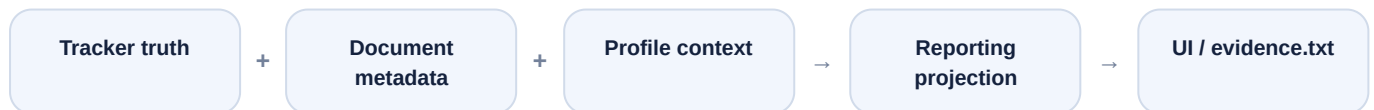
Tracker applications/events, Store document metadata and Profile data through owner-scoped service APIs.

## Read models

Application totals/status/date buckets, recent content-free activity, indicative time and text evidence export.

## Privacy

Document activity reports version/type/lifecycle—not CV or letter content. Empty, small and rich histories have explicit behaviour.



**Indicative means indicative.** Time estimates are explanatory product aids, not payroll, benefits or regulatory calculations.

Evidence: Reporting OpenAPI 2.1, summary/journal/evidence controllers, document-activity PR #21 and central reporting journey.

# Deterministic personas are engineering infrastructure

## What it provides

- ✓ versioned synthetic candidate personas;
- ✓ provider responses and error states;
- ✓ named reset/seed/verify lifecycle;
- ✓ service-owned cleanup orchestration;
- ✓ stable model fixture responses;
- ✓ guarded non-production APIs.

## What it cannot prove

- current provider uptime or changing advert shape;
- real model quality, latency or cost;
- rich real-person evidence distribution;
- production security/operations;
- employment outcomes.

## Unsafe testing risk

Captured personal/provider data, accidental live calls, paid model spend and non-repeatable failures.



## Governed test state

Synthetic datasets, explicit provider modes, authenticated owner reset and deterministic verification.

## Why canonical personas remain

Minimal, typical, rich, stress, uploaded-CV, manual-first and career-changer profiles exercise different structures while staying source-controlled and non-personal. Bernard's persistent real account remains a separate runtime user and is never a fixture dependency.

---

# 7

canonical persona shapes

---

# 0

paid provider calls in fixture regression

---

# 1

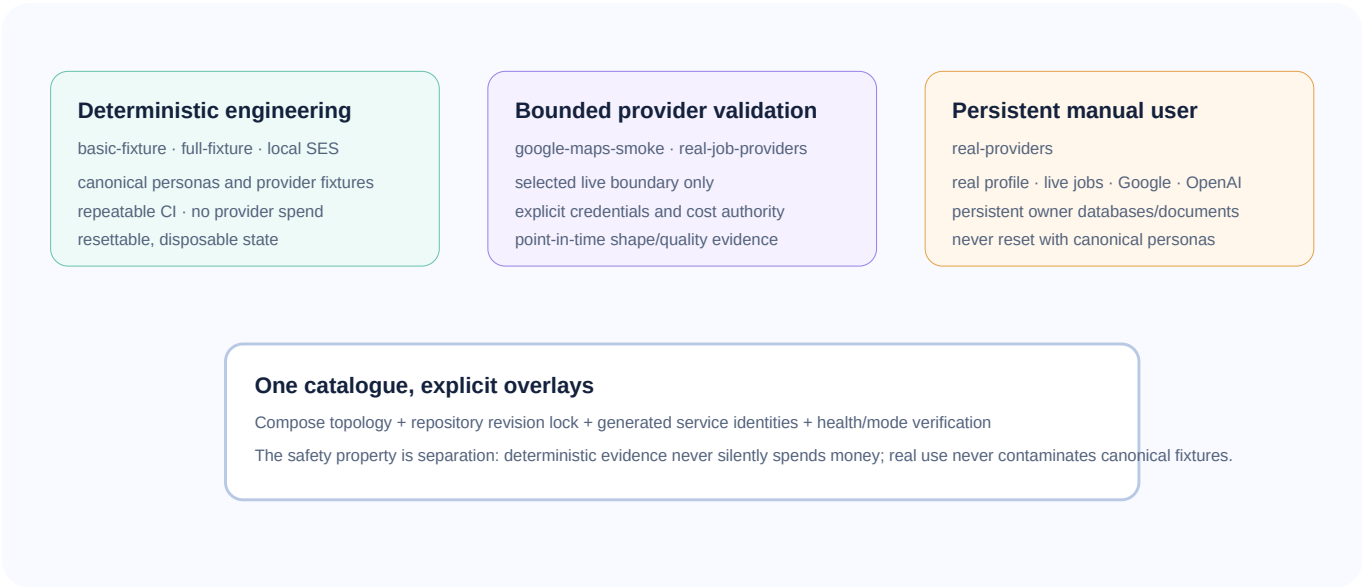
named-state owner for orchestration

---

# strict

non-production safety boundary

# Separate confidence questions require separate modes

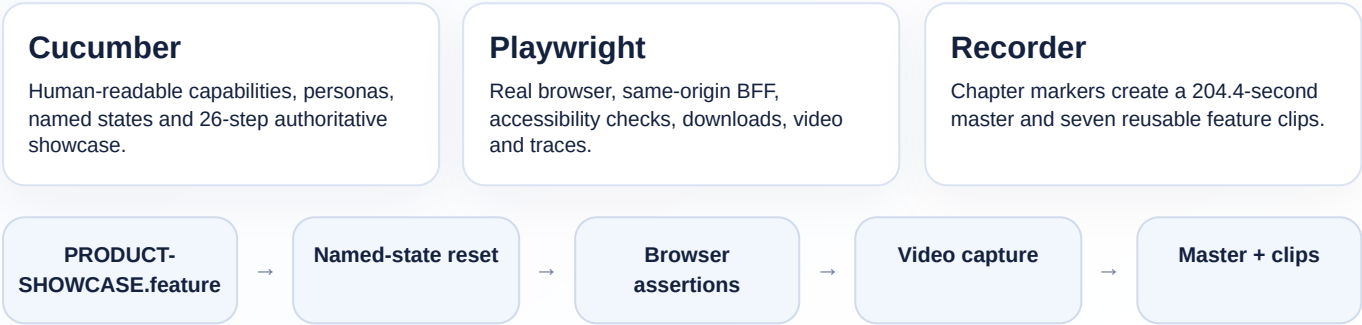


| Profile            | Purpose                                                 | External state                    |
|--------------------|---------------------------------------------------------|-----------------------------------|
| basic-fixture      | Account/profile/location/search fast path               | Deterministic                     |
| full-fixture       | Complete regression/product showcase                    | Deterministic                     |
| full-local-ses     | Production email adapter against LocalStack             | Local deterministic               |
| google-maps-smoke  | Only Google live                                        | Explicit bounded live             |
| real-job-providers | Five jobs + Postcodes live; LLM/Stripe fixture          | Explicit live jobs                |
| real-providers     | Live jobs, Postcodes, Google and OpenAI; Stripe fixture | Persistent authorised manual user |

**Current evidence.** The final health check reported 28 ready application services. Persistent PostgreSQL and document volumes survived ordinary restarts; secrets and personal data remain outside Git.

# The showcase is an executable product journey

```
Scenario: Alex moves from a new account to a tailored application and interview
  Given the showcase candidate is ready to register
  When he opens Job Seeker Copilot and registers an account
  And Alex builds a rich professional profile and evidence library
  And Alex discovers the selected showcase job
  And Alex generates a CV and cover letter for the same showcase job
  And Alex reviews both generated documents
  And Alex progresses the same application to interview
  Then profile, documents, application and reporting state persist after sign-in
```



**Disclosure.** Promotional recordings use deterministic System Data. That is a strength for repeatability, but they are labelled fixture-backed and never used as live-provider proof.

# Each layer answers a different confidence question

| Domain           | Unit/component                    | Integration/contract               | E2E/live evidence                         |
|------------------|-----------------------------------|------------------------------------|-------------------------------------------|
| Identity/profile | validation, revisions, cookies    | PostgreSQL, JWKS, owner denial     | register/profile/reload/return            |
| Job discovery    | adapters, mapping, dedupe         | producer clients, partial failure  | five-source fixture; bounded live samples |
| Applications     | transitions, idempotency          | Store reference verification       | save, select, apply, interview, offer     |
| Documents        | lifecycle, MIME, renderer         | PostgreSQL/filesystem/ClamAV       | upload/generate/history/download          |
| AI generation    | parser, repair, claims, fallback  | wallet/store/tracker recovery      | fixture matrix + bounded paid samples     |
| Location         | canonical mapping, session bounds | Postcodes/Google clients           | fixture commute + live Google smoke       |
| Reporting        | buckets, activity, redaction      | owner-scoped upstream clients      | empty/rich history and UI                 |
| Infrastructure   | catalogue/policy/render tests     | clean build and health/mode checks | capacity and restart persistence          |

32/32

Cucumber scenarios

244/244

journey steps

460/460

Angular client tests in closure evidence

248/248

CV/letter domain tests in closure evidence

Proves quickly

Validation, mapping and deterministic edge behaviour.

Proves together

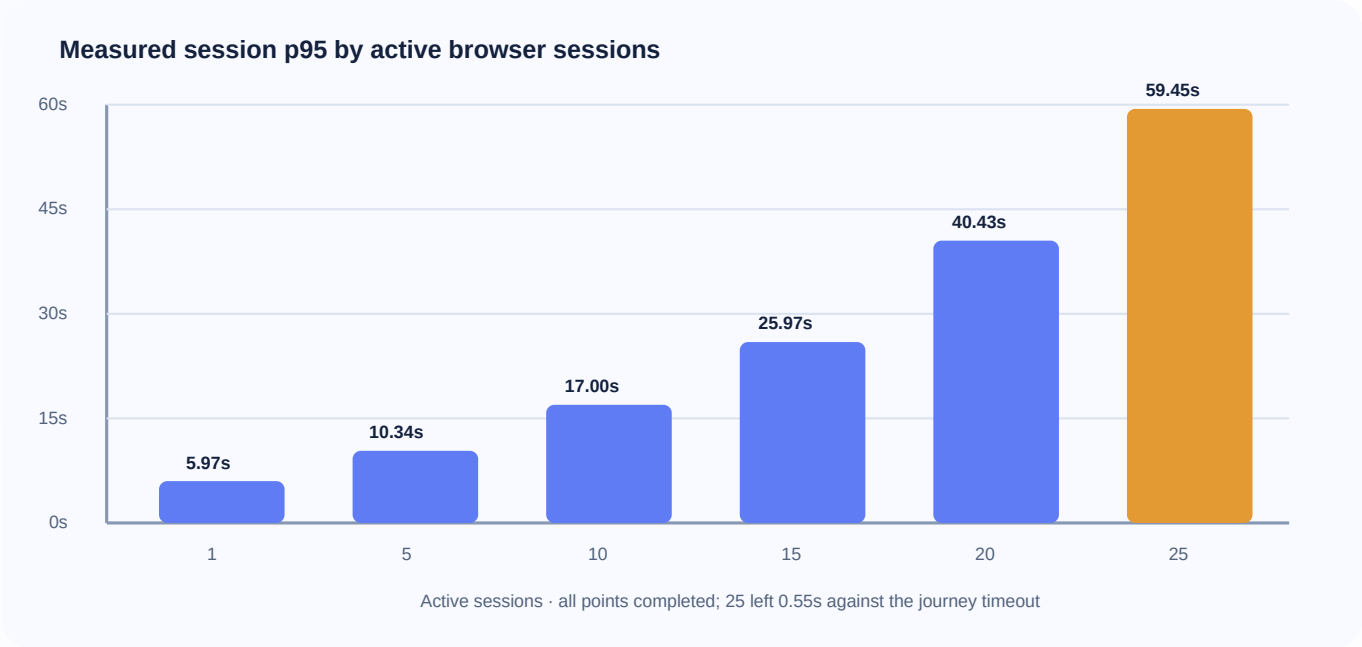
Schema compatibility, persistence and multi-service ownership.

Proves realistically

User comprehension, live shapes, rich data and dated workload behaviour.

No count is universal confidence. Retained totals identify a dated closure run; the dossier does not imply every repository test was re-run during document authoring.

# A failed benchmark found a product defect before it found a server limit

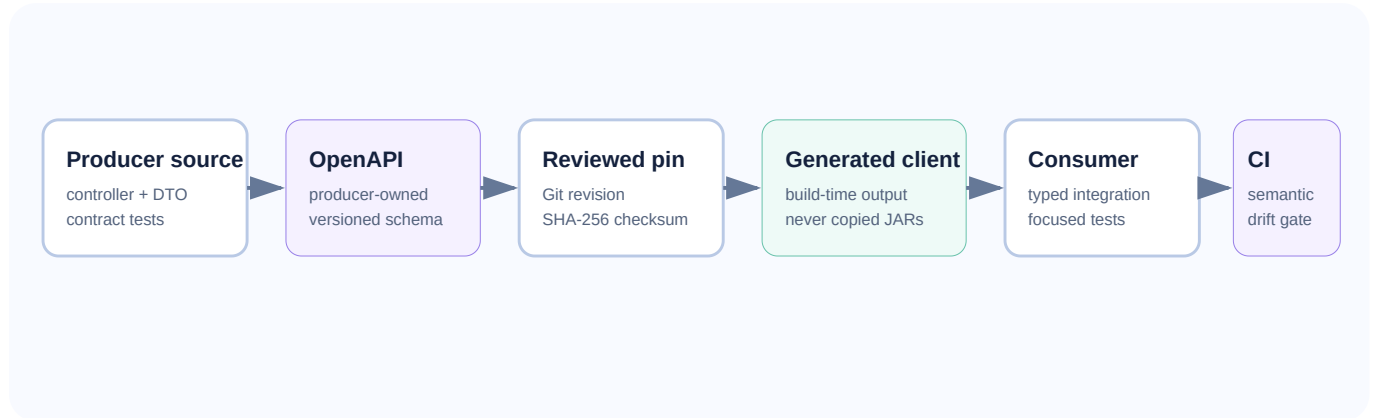


| Sessions | Completed | Session p95 | Response p95 | Peak Docker memory |
|----------|-----------|-------------|--------------|--------------------|
| 1        | 1/1       | 5.97 s      | 182 ms       | 8.01 GiB           |
| 5        | 5/5       | 10.34 s     | 571 ms       | 8.14 GiB           |
| 10       | 10/10     | 17.00 s     | 709 ms       | 8.35 GiB           |
| 15       | 15/15     | 25.97 s     | 472 ms       | 8.42 GiB           |
| 20       | 20/20     | 40.43 s     | 609 ms       | 8.66 GiB           |
| 25       | 25/25     | 59.45 s     | 546 ms       | 8.64 GiB           |

**0/25 → 25/25.** The original failure was not an application-service crash. A refreshed immutable result gained a canonical ID, causing Angular to recreate the card during interaction. Stable provider-job tracking fixed the browser identity race.

**Why no 50.** Throughput peaked at 10; 25 left 0.55 seconds against the 60-second journey deadline. The cautious local operating point is 15. Browser workers were outside Docker measurements; AWS capacity remains unproven.

# Independent repositories require explicit compatibility discipline



## Producer ownership

Controller/DTO source and contract tests produce the reviewed OpenAPI. Consumers do not edit private copies.

## Revision + checksum

Consumer metadata names the producer Git revision and schema SHA-256, making drift reviewable.

## Generated clients

Maven/TypeScript clients are disposable build output. Checked-in local JAR coupling was removed.

## Delivery engineering

| Concern                | Practice                                                            | Failure prevented                      |
|------------------------|---------------------------------------------------------------------|----------------------------------------|
| Branches               | Feature branch → PR → green CI → develop; main separately protected | Unreviewed integration changes         |
| Parity                 | Workspace lock and catalogue coordinate reviewed revisions          | "Works only in my checkout"            |
| Semantic compatibility | Policy tests inspect required routes/schemas/enums                  | Compiles but changes meaning           |
| Security/build gates   | Secret history, dependency, licence, clean clone/container checks   | False-green or irreproducible releases |

**Architecture consequence.** Contract mismatch bugs—claim ledger bounds, timestamp offsets, application provenance and generated client metadata—became explicit PRs and regression gates rather than runtime surprises.

# Failures that changed ownership or workflow

- 1 Generated document not reliably associated with its application.**  
Found by realistic E2E. Generation linked selections before authoritative application identity/reconciliation was ready. The flow now creates/relinks the application and verifies exact references before selection; fixture and E2E preserve regression coverage.
- 2 Unsupported identity and claims could appear fluent.**  
Prompt-only trust was insufficient. Trusted identity moved outside the prompt; strict claim ledgers and evidence-path validation now reject unsupported specific claims.
- 3 Payment browser proxy routing could imply a capability that was not ready.**  
The BFF now fails closed for beta-disabled payment routes while backend wallet/fixture work remains testable and documented separately.
- 4 Capacity refresh destroyed the job card under interaction.**  
Playwright concurrency exposed unstable Angular tracking when canonical identity appeared after refresh. Stable provider-job identity changed 0/25 to 25/25.
- 5 Exact duplicate model structure caused full operation failure.**  
Narrow duplicate repair was added, followed by the same claim validation. Semantic rewrites remain forbidden.
- 6 Provider dates and offsets did not share one meaning.**  
Compatibility normalisation was added at provider/domain boundaries; open backlog items still guard offset preservation where contracts remain weaker.

Each fix added regression protection at the lowest useful layer and, where user behaviour mattered, at browser/E2E level.

# Failures that changed recovery, quality or boundaries

- 7 Rich-profile evidence was omitted or poorly prioritised.**  
Compact personas hid realistic selection pressure. Purpose-sized prompts, richer role/project evidence and deterministic relevance selection now preserve important software experience.
- 8 Model output budget and operation deadline disagreed.**  
A larger CV budget alone was not enough. LLM bounds, gateway read deadlines and durable absolute deadlines were aligned; ambiguous outcomes enter reconciliation rather than blind retry.
- 9 Selected-output recovery expected both documents.**  
After CV-only/letter-only selection shipped, recovery could misread missing unselected references. Operation state and client UI now report partial success truthfully.
- 10 Google location did not fit the direct postcode gateway design.**  
A Location Service was extracted; Google Maps Gateway now owns provider key/DTOs while canonical persistable fields stay provider-neutral.
- 11 Professional export work was stranded outside the integrated branch.**  
Git history located and reconciled template behaviour. Real rich-profile inspection then fixed A4 hierarchy, page breaks and pagination.
- 12 Contract/client drift recurred across independently versioned services.**  
Producer OpenAPI, reviewed revisions/checksums, generated clients and semantic policy gates replaced copied binaries and undocumented shape assumptions.

**Maturity signal.** The important result is not that defects existed; it is that realistic evidence identified why earlier tests missed them and converted them into architecture and regression controls.

# Implemented controls with explicit assurance limits

**Identity**

HttpOnly cookies, CSRF, verified subjects and route-scoped service tokens.

**Lineage**

Digests and stable IDs across evidence, jobs, claims, documents and applications.

**Untrusted content**

Provider text, uploads and model output cross validation/allowlist/scanner boundaries.

**Personal data**

Real profile/CV/contact data remain in owner databases and outside fixtures, logs and Git.

**Secrets**

Runtime-only environment mechanism; provider keys remain in owned gateways.

**Wallet**

Reservations, operation keys, idempotent transitions and unknown-outcome checks.

## Implemented, disabled, limited and post-beta

| Classification                | Examples                                                                                                                                   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ● Implemented/demonstrated    | Profile/evidence, five-source discovery, applications, document generation/export, reporting, Google autocomplete, persistent local state. |
| ● Implemented but not exposed | Live-capable Stripe checkout/webhook boundary; browser purchasing remains disabled.                                                        |
| ● Known beta limits           | No semantic suitability score; bounded provider/model samples; manual accessibility and production security assurance incomplete.          |
| ● Operational work            | AWS application deployment/workload validation, HA/backups/monitoring, larger concurrency and first-user operations.                       |
| ● Post-beta product           | CV-to-profile import, richer matching, submission confirmation and application-answer tooling.                                             |

**No compliance claim.** These controls demonstrate design intent and implemented boundaries—not certification, complete penetration testing, production HA or regulatory approval.

# Substantial engineered product, candid controlled beta

## Working/demonstrated

- ✓ Authenticated persistent profiles and revisioned evidence
- ✓ Five-source fixture and bounded live discovery
- ✓ Canonicalisation, dedupe and immutable saved jobs
- ✓ Applications, exact document selections and history
- ✓ Grounded CV/letter generation, recovery and export
- ✓ Google-assisted location and advisory commute
- ✓ Reporting, System Data, E2E and promo automation
- ✓ 28-service persistent manual environment

## Still bounded/unproven

- No semantic candidate/job score
- No live browser Stripe purchases
- No AWS application workload validation
- No production uptime, HA or user-scale claim
- No full manual accessibility/security assurance
- No user, revenue or employment-outcome evidence

## What a technical interviewer can discuss

Java/Spring boundaries

Angular/BFF security

OpenAPI contracts

PostgreSQL ownership

Provider normalisation

Idempotency

LLM grounding

Failure recostitution

Document lineage

Playwright/Cucumber

Capacity diagnosis

Docker environments

GitHub delivery

Product decisions

This was not a UI wrapped around an LLM. It was iterative engineering across contracts, state, providers, documents, failure recovery, testing and real-world correction.

The strongest evidence is the ability to recognise when the system's earlier confidence was too narrow, reproduce the failure, identify the ownership error and improve both architecture and tests.

Pages 40–42 provide the complete repository catalogue. Contact details and job-specific suitability remain in the accompanying CV and cover letter.

# Client, identity, profile, location and discovery

## Client

### job-seeker-copilot-client

**Angular 21/TypeScript/Express.** UI and same-origin BFF; cookies only. Calls identity, jobs, location, documents and reporting. Allowlisted routes, CSRF, safe provider content and truthful partial states.

## Identity

### authentication-service

**Spring Boot 4.1/PostgreSQL.** Accounts, credentials, refresh/session lifecycle, RS256/JWKS. Called through UMG; optional SES. Canonical email and bounded login behaviour.

## Identity facade

### user-management-gateway

**Spring Boot 3.5.** Browser auth/profile/evidence facade. Stateless; derives trusted identity and maps stable redacted errors.

## Profile/evidence

### user-profile-service

**Spring Boot 4.1/PostgreSQL.** Profile, preferences, evidence entries/revisions and snapshots. Optimistic revisioning, confirmation, provenance and digests.

## Location facade

### location-gateway

**Spring Boot 4.1.** Browser v1 compatibility and v2 autocomplete/resolve. Calls Location Service; bounded cache/rate/error state.

## Location domain

### location-service

**Spring Boot 4.1.** Provider-neutral canonical resolution and commute orchestration. Transient suggestion sessions; calls Google/Postcodes gateways.

## Location provider

### postcode-io-gateway

**Spring Boot 4.1.** Postcodes.io mapping with fixture/live mode, cache, timeout, retry, circuit and safe URI construction.

## Location provider

### google-maps-gateway

**Spring Boot 4.1.** Sole Google key/DTO/attribution boundary for Places and Routes; disabled by default; transient bounded state.

## Job facade

### job-finder-gateway

**Spring Boot 3.2.** JWT-protected search, save and application facade. Calls Job, Profile/UMG and Tracker; preserves provider status and provenance.

## Job domain

### job-service

**Spring Boot 3.2/PostgreSQL.** Five-source fan-out, canonical mapping, dedupe, detail hydration and immutable saved jobs.

## Enrichment

### job-matching-service

**Spring Boot 3.2.** Application reconciliation and advisory commute metadata. Stateless; no semantic suitability score.

# Providers, applications, documents and AI

## Provider

### reed-gateway

Reed key, search/detail mapping and fixture/live mode. Stateless; source/apply identity and bounded failures.

## Provider

### adzuna-gateway

Adzuna app credentials, pagination and mapping. Stateless; workplace type and provenance retained.

## Provider

### jsearch-gateway

RapidAPI JSearch authentication/mapping. Stateless; variable aggregated IDs, dates and descriptions.

## Specialist provider

### nhs-jobs-gateway

Official NHS self-serve XML adverts and deterministic fixture; authoritative source identity flows into applications/reporting.

## Specialist provider

### apprenticeships-gateway

DfE Display Advert API v2; authenticated acquisition to an atomic bounded vacancy snapshot.

## Applications

### application-tracker-service

**PostgreSQL.** Applications, transitions, immutable events and exact/frozen document slots; Store verifies references independently.

## Orchestration

### document-generation-gateway

**PostgreSQL operation ledger.** Snapshots, reservations, generation, approval, export, upload, application link and recovery checkpoints.

## AI domain

### cv-cover-letter-service

Prompting, typed parsing, narrow repair, grounding, claim ledger and deterministic CV fallback; optional encrypted rejected-output quarantine.

## AI provider

### llm-gateway

OpenAI/fixture transport, admission and deadline bounds, typed usage evidence, classified retry and audit.

## Documents

### document-store-service

**PostgreSQL + object provider.** Families, immutable versions, pointers, lineage, file bytes, ClamAV boundary and retention lifecycle.

## Export

### document-export-service

Stateless Apache POI/OpenPDF DOCX/PDF renderer and bounded DOCX ingestion; stable replay-safe keys.

# Reporting, payments, testing, infrastructure and landing

## Reporting facade

### reporting-gateway

JWT-protected summary, journal and evidence-text routes; stateless bridge to Reporting Service.

## Reporting

### reporting-service

Read-only owner-scoped application/profile/document projections; no domain writes or authoritative DB.

## Wallet

### payment-service

**PostgreSQL.** Wallet, append-only ledger, reservations, commit/release, expiry and recovery reconciliation.

## Payment facade

### payment-gateway

Wallet/pricing/checkout facade; backend fixture composed while browser checkout remains deliberately unavailable.

## Payment provider

### stripe-gateway

Checkout/session and signed-webhook boundary; live-capable code, fixture standard profile, calls Payment Service.

## Test data

### system-data-service

Governed synthetic personas, provider fixtures and guarded named-state lifecycle; strictly non-production.

## E2E

### e2e

Playwright/Cucumber browser journeys, accessibility, capacity and executable promotional recording.

## Runtime/delivery

### infrastructure

Compose topology, catalogue, profiles, secrets, revision locks, bootstrap, health/mode checks, capacity and documentation.

## Public site

### job-seeker-copilot-landing

Separate Angular/AWS SAM waitlist/contact stack using API Gateway, Lambda, DynamoDB and SES with independent launch gates.

The catalogue describes ownership, not organisational ceremony. The engineering value is that changing one domain—model provider, document storage or job source—does not silently redefine every other domain.

## Accompanying evidence

This detailed dossier is supplied alongside a job-specific CV and cover letter. The 204.4-second deterministic showcase and seven feature clips demonstrate the product journey. Private source/PR evidence can be discussed in technical interview or reviewed by arrangement.

Contact details are provided in the accompanying CV.